Geosocial Platforms as Media of Urban Policy Mobility [DRAFT 2]

Khalil J. Martin[1]

1. University of Toronto, Department of Sociology

Abstract

Geosocial platforms (GP) like Yelp, TripAdvisor, and Google Local Guides are reshaping how notions of place form and are realized, yet their role in urban policy mobility remains underspecified. This paper presents three empirical studies set in and around Philadelphia to examine how GPs intervene in processes conventionally associated with urban policy mobility. The first uses TripAdvisor data and topic modelling to assess traveller responses to the reconstruction of LOVE Park, illustrating GPs as public feedback tools. The second uses spatial analysis of Yelp review trajectories to show how the platform's 'first review' metric systematically shapes users' urban movement patterns, effectively embedding its own form of urban policy. The third uses transition matric modelling and LLM classification of Google Local reviews to explore the inter-municipal flow of brewpub sentiments and improvement suggestions across Pennsylvania. Together, the three short studies reframe urban policy mobility around the more general concept of place-based influence, shifting analytical attention from bounded administrative territories and urban professionals toward global computational infrastructures and their distributed users.

Introduction

How do geosocial platforms such as Yelp, TripAdvisor, and Google Local Guides, intervene in processes associated with urban policy mobility? Urban policy mobility scholarship has sought to understand the role of the internet and other digital technologies in the generation, appreciation, and circulation of urban policies and place-based ideas. This work often follows professionals such as planners, architects, and engineers, as they generate, debate, rank, and index ideas which can be adopted and administered by government officials and staff (Ward, 2024; Silva and Ward, 2024). However, the radical changes digital mediation have made to practices of urban ideation, appreciation, and circulation over the past 10 to 20 years remains underspecified in urban scholarship (ibid.).

A major form of digital transformation has come through development and adoption of peer-to-peer (P2P) platforms. A growing platform ecosystem has forced urban scholars to question conventional thinking at the intersection of urban, policy, and mobility (Barns, 2019). There is a general understanding among urban scholars that, through their widespread use, platforms gather up and manipulate urban life in ways that are often at cross-purposes with local governments and communities, if not simply ignorant of their territorial concerns (eds. Hodson et al., 2021; eds. Strüver and Bauriedl, 2022). Overall, *whereas the conventional approach to urban policy mobility centers territorial administrations and policy professionals, P2P platforms shift the emphasis towards global computational infrastructures and their mobile users*.

This paper specially addresses geosocial platforms (GPs). These are platforms such TripAdvisor, Yelp, and Google Local Guides, which allow users to peruse amenity maps, publish amenity reviews, interact with each others' reviews, and more. Through three short empirical studies, this paper explores three specific ways in which GPs may intervene in processes associated with urban policy mobility, with a focus on cultural activities. The first explores GPs as tools for engaging public feedback about place-based interventions such as the reconstruction of public spaces. The second study explores GPs as computational infrastructures which embed their own kind of urban policy, specifically through the design of user metrics and rewards which influence urban practice in certain ways. The third study explores GPs as facilitators of the sentiment and idea flow across broad amenity landscapes, specifically by enabling and promoting the reviews of users who travel around to specialize in one amenity category or another.

While the first study does not necessarily challenge conventional urban policy mobility thinking, the latter two studies do. In the second study, the platform itself, rather than government bureaucracy and staff, becomes the becomes the structure which implements policy. And in the third study, GP users, rather than policy professionals, are cast as the main mobilizers of place. In sequence, the studies turn over conventional thinking at the intersection of urban, policy, and mobility, using GPs a conceptual pivot: turning from a focus on official territories and professional experts to a focus on ubiquitous computing and prolific users. At the same time, each study represents a concrete avenue of research on the influence of platforms within urban cultural life. Because the paper includes three different empirical studies, and space is limited, discussions of previous work are woven into the methodologies and study sections where relevant. The code and data used in the studies are publicly available (https://github.com/kjrm/Geosocial_Platforms_as_Urban_Policy_Mobility_Media-Replication.git) for replication or adaptation.

Approaches and Methodologies

Study 1: Platform as feedback tool

The first study approaches GPs as tools for engaging public feedback about place-based interventions. When authorities intervene in the urban landscape by creating or altering an amenity, the reviews posted by GP users about this amenity may be processed to help assess the efficacy of intervention. This approach to GPs aligns closely with efforts to use GPs as data sources for data-driven urban policy (Arribas-Bel et al., 2016; Glaeser et al., 2019; Song et al., 2021).

In this first study, the intervention in focus is the reconstruction of Philadelphia's JFK Plaza aka LOVE Park, and the GP in question is TripAdvisor.

TripAdvisor posts about LOVE Park before, during, and after reconstruction are processed using topic modelling to help appreciate the transformation of visitor experiences precipitated by the intervention. The section further explores the reconstruction of LOVE Park, and the subsequent creation of LOVE Malmö in Sweden, as an example of 'negative idea mobility', where place-based ideas are mobilized in response to loss, in the case the loss of the skateboarding mecca that LOVE Park once was.

Study 2: Platform as policy implement

The second study approaches GPs as policy implements themselves. That is, GPs may embed different 'urban policies' in their variable designs of user affordance, metrics, algorithms. If GP design decisions influence urban practice in systematic ways, then they can be said to implement their own kind of urban policy. This approach to GPs aligns closely with critical views about the governmentality of digitally-mediated landscapes (Klauser et al., 2014; Dammann et a., 2022).

This second study explores the influence of the Yelp firsts metric (where users are rewarded for being the first to review an amenity) on Yelp review trip trajectories in Philadelphia metro area. First, it conducts a tail analysis on the user distribution of firsts to identify users who likely actively pursue firsts (scout) versus one who don't (non-scout), by measuring this distribution against an expected alternative distribution where users just so happened to be the first to review every now and again. Then, it matches each scout with a non-scout control of equal number of reviews. Finally, it compares the review trajectories of scouts versus non-scouts to test if the former are more geographically dynamic than the latter, as this supports the idea that the platform metric systemically influences spatial practices and, therefore, that the platform implements its own kind of 'urban policy' among the citizenry of reviews.

Study 3: Platform as facilitator of sentiment and idea flow

The third study approaches GPs as facilitators of sentiment and idea flow across broad landscapes. On one hand, GPs make a broader range of amenity locations more legible and accessible to prospective visitors, by publicly organizing user amenity reviews, and thereby making amenity providers more accountable to visitor experiences. On the other hand, GPs provide each user with a platform to catalogue and develop taste, making their own experiences more recountable, as part of a publicly developing profile. By elevating provider accountability, and enabling user profiling, GPs may present enhanced conditions for urban 'cross-pollination', where each amenity is a flower, every reviewer is a pollinator, and the sentiments and ideas contained in those reviews are the pollen. This approach to GPs is in the vein of urban scholarship which views the internet as a medium for the comparative appraisal of places (White and Kitchin, 2021; Ward, 2024), except here we are casting GP users as the appraisers and considering a finer-grained notion of place (Zukin et al., 2017).

This third study explores the flow of brewpub ideas across the state of Pennsylvania, through Google Local Guides reviews. It traces every time a user crosses municipality lines between brewpub reviews and tallies these transitions directionally. These directional transitions are then mapped to appreciate the flows of brewpub sentiments among Pennsylvania municipalities. Next, using LLM labelling methods, it considers only brewpub reviews posted by users who have made at least one explicit suggestion for improvement among their non-five-star brewpub reviews, and, further, only those reviews that have received a public response from the amenity provider (strong indication of reception). The geographic distribution of these remaining transitions is analysed to appreciate the flow of brewpub ideas facilitated by the platform. Overall, this last study explores how GPs may be instrumental in the regional development of place-based industries such as brewpubs.

Study 1: Platform as feedback tool

Reconstructing LOVE Park

JFK Plaza was constructed in 1965, under executive direction of the Philadelphia Planning Commission, to serve as a civic forecourt for Philadelphia City Hall. The plaza featured granite surfaces and steps, minimal greenery, and benches around a large fountain. During the 1979 U.S. bicentennial celebration held at the plaza, Robert Indiana's pop art sculpture "LOVE" was presented. Due to the sculpture's popularity, it was installed on a permanent foundation and became a cherished icon of the 'City of Brotherly Love'. Over time, JFK Plaza grew to be more-affectionately called LOVE Park (Mires, 2018).

In the 1980-90s LOVE Park became a world-renowned skateboarding destination due to its expansive design and hard, curving surfaces. In 2001, X Games held its first Skateboard Street competition in the area encompassing LOVE Park, and around City Hall, with the conspicuous cooperation of City officials (X Games, 2021). The event was a concession around what had already become a site of conflict in the years prior. Youthful crowds of skateboarders and onlookers would gather nightly to participate in skate culture and more, which many experienced as disruptive. Tensions rose between authorities and skaters, through sporadically-enforced skating bans. LOVE Park at the turn of the millennium was a symbol of conflict between youth culture and civic life (Borden, 2019).

The City decided to reconstruct LOVE Park in 2014. The intention was not just to repair aging infrastructure, but to create a public space more hospitable to a broader range of visitors, which can generate revenue, and garner a more polished global reputation. Mayor McNutter expressed that the new plaza would be one "they are going to be talking about" alongside the likes of "Bryant Park in New York" (MacDonald, 2014). The succeeding Mayor Kenny acknowledged the significance of LOVE Park as a skateboarding destination, but emphasized the plaza's role as a ""gathering place for Philadelphians young and old, for neighbours, business people and tourists, and for individuals seeking to come together to share civic pride" (City of Philadelphia, 2016). As a minor concession, he invited skateboarders to use the plaza up until its closure date on February 15th, 2016 (ibid.). In place of the hard-surfaced space with a large foundation, the new plaza would be softer, featuring more of a splash area than a fountain, more lawn space, and a visitor center.

Although the plaza may draw visitors for many reasons, the LOVE sculpture remains the broadly-recognized element. During the plaza's reconstruction, the City's Office of Arts, Culture, and Creative Economy (OACCE) oversaw the sculpture's restoration, managed its temporary relocation to a location outside City Hall, and co-hosted a Valentine's Day parade celebrating it's return in February 2018. Given the global ambitions of LOVE Park, first as a skating mecca, and then as a more tourist-friendly stop, the City may have been interested to know how travellers experienced the plaza's renovation. As TripAdvisor is the most popular GP which caters heavily to global travellers, the reviews posted on the platform may serve as one relevant source of feedback on this policy-driven urban intervention (Song et al., 2021). Can TripAdvisor reviews of JFK Plaza be used to help assess how well the reconstruction was managed, and how well the overall redesign was received among travellers, at least according to the City's perspective and interests?

TripAdvisors before, during, and after reconstruction

TripAdvisor posts of JFK Plaza were collected for all dates up until the end of February 2020. This includes 642 posts, including review text (100%), rating (100%), month-year of experience (90%), and date written (100%). The records are split into three periods—*before*, *during*, and *after* reconstruction—according to the month-year of experience. Records not specifying a month-year of experience were dropped. Because only the month-year of experience is specified and not the precise date, records where the month-year of experience coincide with the closing and reopening ('buffer') months were also dropped. This leaves 568 (88%) records, with 339 records in the before period, 114 in the during period, and 115 in the after period.

Descriptive statistics show that there are a greater proportion of 5/5 ratings after versus before reconstruction (0.22 → 0.32). During reconstruction, the proportion of 1/5 ratings increased (0.04→0.11) as expected, but so did the proportion of 5/5 ratings (0.22→0.27). This bifurcation could reflect a somewhat successful rerouting of visitors from the cordoned-off plaza to the temporarily-relocated LOVE sculpture. Several reviews suggest that that is the case, and some posts even include photos of the wayfinding to help other visitors find the sculpture. See Section A in Supplemental Material for more descriptive statistics.

Expressions of a changing scenery

To appreciate how TripAdvisor experiences around JFK Plaza changed across the three periods, period-specific BERTopic modelling was conducted on the review texts. BERTopic is a transformer-based method that combines contextual sentence embeddings with density-based clustering to identify informative themes in corpora of short texts (Egger and Yu, 2022). Rather than imposing topic continuity across periods, each period is treated as a distinct discursive context. For simplicity, the same model parameters were used across all three periods, and these were selected to prioritize interpretability and comparability across periods (See Section A in Supplemental Material for model specification). See Supplemental Material Table A-2 for the topics imputed across the three periods. As expected, most topics features relatively generic references to the location and picture-taking. However, at least three or four topics seem comparatively noteworthy.

Topics 1-5, 1-6 and 1-7 gesture to potentially negative experiences with skateboarders and people living rough around the plaza, in the period before reconstruction. “Homeless people sleeping on benches,” reads one Topic 1-5 review, “We left the park feeling uncomfortable” (reviewer before reconstruction). One Topic 1-6 review reads: “we were basically harassed by some homeless dude to let him take our picture. He wouldn’t let us say no, so we caved” (Reviewer before reconstruction). Other before-reconstruction reviews describe similar experiences. A Topic 1-7 review reads: “This park is inhabited by homeless people and vagrants. This is not a place that tourists should go after the sun goes down” (Reviewer before reconstruction). “Was really surprised how scary this park was,” reads another (Reviewer before reconstruction).

The other ‘social type’ of the plaza, in addition to “homeless people”, were skateboarders. As a Topic 1-6 review reads, dramatically: “Cesspool. Homeless people everywhere. Skateboarders everywhere who run you over and fling their urine soaked boards at innocent bystanders” (reviewer before reconstruction). However, several reviews in this topic mourn the state of what was once a great skating destination. They celebrate skaters while lamenting the plaza’s neglect. For example, one Topic 1-6 review reads in full:

> “*Growing up in the 80’s and 90’s skateboarding I was excited to finally go see the iconic Love Park. While it still looks the same as remember from the videos it feels dead. When I was young I would have done anything to go visit this park and watch and feel the electricity in the area. Now its just a area for homeless people and their waste. Shame on the city for letting this happen. They had a chance to put skateboard guidelines in place to keep the youth in the area involved and keep the area alive. Now it is scary at night and scary in the day as well. Hopefully the next time my family goes my kids can skate hassle free. Come on Philadelphia either clean*

it up right or give it to a group of young people who will take care of it" (Reviewer before reconstruction).

Other Topic 1-6 reviews express similar sentiments. On the one hand: "As a skateboarder on vacation in Philly, I knew I had to visit one of skateboardings most iconic skate spots". On the other: "many homeless, sketchy people eyeing me, asking for money to take a photo. Stay out of this shady place if you can" (Reviewer before reconstruction).

The themes describing a neglected skating site do not appear in the after-reconstruction period. Instead, as one Topic 3-5 review reads, "There were kids playing in the fountains and parents and young couples milling about" (Reviewer after reconstruction). The reconstructed plaza is a place where one is proud to exclaim, "We got engaged at Love Park!!" (Reviewer after reconstruction).

From Philadelphia to Malmö, with loss

One outlier review is worth highlighting, not just for its heartfelt language of loss in the after-reconstruction period, but also because it is the only post observed to have received a response, the very next day, presumably from someone employed by the City. "History reveals the joy of skateboarding was once had here", writes the prolific reviewer, and further, "the park of yesteryear is no more now replaced with a flat, sterile lifeless designed slab of concrete" (Reviewer after reconstruction). The responder informs the reviewer of other skating infrastructure in the city, and highlights some of the plaza's new features. The response suggests that the platform *was* being monitored for public feedback.

The redesign of LOVE Park triggered a sort of 'negative idea mobility', in which a place was mobilized in response to loss rather than success (see Lovell, 2017). Once a globally iconic street-skating site, LOVE Park's transformed design and management significantly reduced its skatability, a transformation widely documented and lamented in international skate media (e.g. Thrasher, Jenkem). Many in transnational skateboarding networks shared the sentiments of the TripAdvisor reviewer above. Over in Malmö, Sweden, where municipal officials had established collaborative commitments to skate-friendly urban policy, this loss became a source of inspiration rather than discouragement (Borden 2019). LOVE Malmö was constructed as a deliberate act of commemoration and translation: a centrally located space that attempts to preserve the architectural language and symbolic aura of the old LOVE Park, using granite from the original plaza (Malmö, 2024). Gustav Eden, who was hired by the city to help promote skating and lead the project, expresses in a *Free Skate Magazine* article that, "When Love Park was being lost, I wanted to try and see if we could help to preserve parts of the space so that the history of Love Park could continue. Give skaters back the opportunity to experience the spot" (Powell 2024). The article describes how city staff, and skate advocates, from both cities interacted to help make this overseas translation possible (ibid.).

Discussion

TripAdvisor reviews suggest a relatively successful transformation of JFK Plaza, from the point of view of generalized hospitality. An important caveat is that TripAdvisor reviewers are not representative of the city's stakeholders. This study merely demonstrates how GPs can be brought into the assessment place-based interventions, and future studies may incorporate multiple sources of information. Further, depending on the scope of future investigations, the methodology may be improved by employing *temporal* topic modelling, instead the *period-specific* topic modelling. Temporal topic modelling tracks how topics evolve, emerge, and disappear over time, which may reveal interesting seasonal dynamics and relationships among topics (see Wang in this issue). Also, whereas the present study used *unsupervised* topic modelling, *semisupervised* topic modelling asks the investigator to predefine topic labels or words to seed the imputation of topics from the corpus. This may produce results that are more relevant to preestablished policy goals or theories (Watanabe and Zhao, 2020).

While we explored how the plaza's transformation was experienced by TripAdvisors, we also traced how it impacted a more specialized, international community in a different way. One result of the reconstruction was a new plaza in another city. Once loss and failure are appreciated as sources of inspiration for place-makers, and not just gain and success (Lovell, 2017), then the materialization of LOVE Malmö from the spolia of LOVE Park fits the conventional mold of urban policy mobility, where professionals from different cities come together to exchange and develop place ideas (e.g. González, 2011; Lee and Hwang, 2012).

However, P2P platforms only amplify the voices of those who use them prolifically, not individuals who may have particularistic attachments to one place or another. They are more than a tool for public feedback; they are reputational systems open to all, which is an added reason why cities may be interested in monitoring them. GP users don't have to be official, long-time residents of a place to make 'discursive investments' which may influence its character by informing material investments (Zukin et al., 2017). This openly influential aspect of GPs is explored in Study 3.

Study 2: Platform as policy implement

Placing first on Yelp

On the Yelp platform, users are awarded a point for being the first to review a listed amenity. The tally of each user's firsts points is displayed on their profile, along with other metrics like the number of votes their reviews have garnered, the number of followers they have, and more. The firsts (hereafter 'first-review') metric is one score around which users can enter playful competition with themselves or with others, where the higher the score the better (Patterson, 2023). If users are thusly motivated to pursue the first review of any listed amenity, does this mean that the platform is systematically shaping urban practices?

Specifically, through the first-review metric, the platform may further encourage users to seek out and try amenities that are lessor known. Although the informational nature of platform participation inherently promotes some such scouting behaviour (Moeller and D'Ambrosio, 2021: 57), the idea is that the first-review metric makes the reward for scouting behavior more acute and credible, especially as it may influence the distribution of tangible perks (Rozek, 2016). It is beneficial to Yelp operators, as well as to everyday information seekers, that some segment of users actively seeks out, and reviews, new amenities. This ensures that the informational coverage of the platform is constantly and promptly extended or refreshed.

However, note that each time an amenity is listed to the platform, this an opportunity for *only one* user to be accredited a first-review point. This is a zero-sum situation which different users may respond to in different ways. Some may choose to actively pursue first-reviews, while others may choose to focus on other metrics. The former effectively serve as the 'scouts' within the swarming collectivity of users. Do these Yelp scouts exist, and does their geographic footprints reflect the role they play? If so, this supports the notion that GPs can embed their own kind of urban policy, as it suggests that a single mutable source (the platform design) systematically influences the conduct of certain individuals within the urban environment.

Identifying Yelp scouts in Philadelphia

This study focuses on Yelp reviewing locations within the Philadelphia metro area. This metro area is defined by drawing a 60km-radius circle around a geolocation (39.9526° N, 75.1652° W) in downtown Philadelphia. From the Yelp open dataset (Yelp, 2025), we select only those reviews for amenities within the circle and posted between 2010 and 2019. The geographic criterion is used to exclude reviews that result from multiday trips or migrations, while also including trips that transgress arbitrary (for our present purposes) municipal boundaries. The temporal criterion is used to avoid the early years of Yelp when first-reviews may have been relatively easy to acquire, as well as the stay-at-home order of 2020 (City of Philadelphia, 2020) and the effects beyond.

The resulting reviews are filtered to get only those reviews posted by users 30 or more total reviews within the spacetime window, enough to establish a pattern of activity. These user reviews are further filtered to include only those users whose review centroid rests within a 15km-radius circle centered at the same geolocation used to the define the study area. This allows us to compare the review trajectories of users who may be operating with a similar home base. Figure 1 shows the study radii.

We know that some users celebrate being the first to review an amenity, for example: "I'm excited to be the first one to review this new cupcake shop in Glendale!" (reviewer in Yelp dataset). However, we want to look beyond these expressions to detect a structural effect of the platform on user trajectories, a consistent deviation from otherwise stochastic expectations (Abbot, 2001). Scouts should have a far higher share of first-reviews that otherwise expected. If users just happened to accumulate first-reviews they more they review, by chance, and there is no competition around first-reviews, then we would expect to the user distribution of first-reviews to be approximately binomial around the overall rate of first-reviews. Through a tail analysis, we can appreciate how different the observed distribution is from this otherwise-expected, binomial distribution. Section B in the online Supplemental Material details this tail analysis. We find that the observed distribution has a fatter, longer tail than the binomial model (see Figure B-1 in Supplemental Material), indicating that many users have a highly disproportionate rate of first-reviews. We use a safe cutoff point along this tail to identify our set of 'scouts'. From our pool of 4,231 users, 297 are thereby identified as scouts. Their average rate of first-reviews is 10.3%, versus 2.2% for the overall pool of users.

Comparing footprints

Now that we have identified scouts, we can compare their geographic footprints to suitable non-scout controls from the same pool. Non-scouts are identified as having no first-reviews. Each scout is matched with a single non-scout with same number of total reviews (yielding 227 pairs), so that we can perform pairwise comparisons of their review-location footprints. For each of the paired users, the total path length, and the radius of gyration, are calculated. The *total path length* captures geographic dynamism. It is calculated by drawing a straight line between each pair of consecutive locations along a path and summing their lengths. Review paths that are more deviating or dynamic are ones that have greater shifts in review location and, therefore, total greater path lengths. The *radius of gyration* captures geographic dispersion (Gonzalez et al., 2008; Pappalardo et al. 2015). It is the root mean square distance of review locations relative the centroid of locations. A more dispersed set of reviews locations should yield a higher radius of gyration. The hypothesis is that scouts with tend to have a higher total paths length, and a higher radius of gyration, than their non-scout peers. Figure 1 shows review location paths for one pair of users with 41 reviews each in the study window.

We find that, in 67.8% of the pairs, the scout's total path length exceeds that of their non-scout control. Taking 50% to be our null hypothesis, a binomial test supports the conclusion that scouts tend to have a higher total path length than non-scouts ($p = 0.0000$), indicating a more dynamic review-location pattern. The average segment length (distance between review locations) for scouts was found to be 10.2km, versus 7.6km for the non-scout controls. A paired t-test supports the notion that scouts tended to travel further distances between review locations than their non-scout counterparts ($p = 0.0000$).

We also find that, in 65.2% of the pairs, the scout's radius of gyration exceeds that of their non-scout control. Again taking 50% to be our null hypothesis, a binomial test supports the conclusion that scouts tend to have a higher radius of gyration than non-scouts ($p = 0.0000$), indicating a more dispersed review-location pattern. The mean radius of gyration for scouts was found to be 10.9km, versus 8.4km for the non-scout controls. A paired t-test supports the notion that scouts tended to have a more dispersed collection of review locations than their non-scout counterparts ($p = 0.0000$).

These conclusions are robust to changes in the study parameters (see Table B-1 in Supplemental Material).

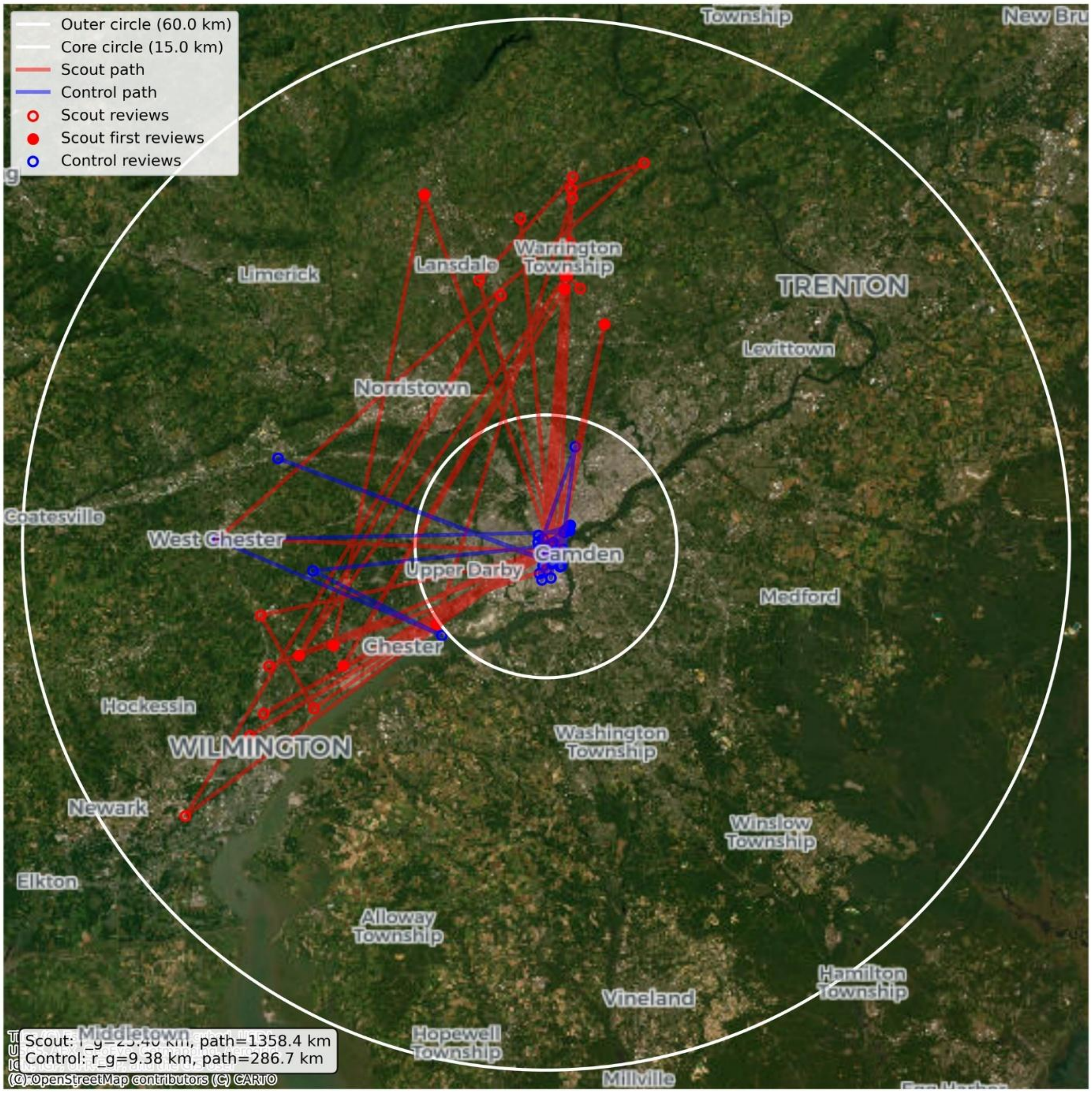

Figure 1: Example review paths for a pair of users, with 41 review locations each. The scout (red) has 9 (22%) first-reviews, a radius of gyration of 25km, and a total path length of 1,358km. Their non-scout control (blue) has no first-reviews, a radius of gyration of 9km, and a total path length of 287km.

Discussion

First, we suggested that Yelp's institutionalization of a first-review metric should structure the observed distribution of users' first-review rates, as some users decide to actively pursue this metric while others choice to focus on other things. Through tail analysis, we found evidence of this structuring. Further, we suggested that the pursuit of first-reviews should have some geographic correlate appropriate to scouting behavior, that users often must 'go out of their way' to experience a first-review. As an urban strategy, this may result in more dynamic and dispersed review locations versus less. Through a pairwise comparisons of the review locations of users with disproportionate high first-review rates (scouts) versus users with no first-reviews (non-scout), we found moderate evidence of this correlation. Overall, this short study points to the structuring effect of platform design on urban practices, giving support to the notion of a platform-embedded urban policy.

Note that, presently, neither TripAdvisor nor Google Local Guides features a similar first-review metric; it is a design choice likely supported by a particular business agenda. This study employed a defensible *within*-platform method for detecting structural effects, but more compelling *cross*-platform methods may be conceivable for future research. By comparing user choices between two or more different platforms, or even between an older and newer design configuration of the same platform, researchers may be able to further illuminate how platform design influences urban practice. While cross-platform compassion is pursued within the orbit of urban studies, this scholarship remains underdeveloped (Törnberg and Söderström, 2025; Xiang et al., 2017) and can profit from the kind of social-computational-statistical reasoning employed in this short study.

Study 3: Platform as facilitator of sentiment and idea flow

Brewpubs across Pennsylvania

Over the past three decades, the craft brewing industry has as one of the most innovative industries in North America (Gatrell et al., 2018). The small-scale entrepreneurship which characterizes the industry has helped breathe life back into postindustrial urban areas, along with amenities like cafes, boutiques, and art spaces (Mathews and Picton, 2023; Apardian et al., 2022). Craft beer is delivered in two categories: (micro)breweries and brewpubs. Breweries produce and distribute beer and sometimes serve a small proportion of beer onsite. Brewpubs produce and serve beer onsite, along with food. In states like California, Pennsylvania, and others, this is a legal distinction which requires different licensing. By integrating beer production into a full onsite experience, brewpubs have helped cultivate local, neo-industrial aesthetics which appeal to the mobile (ibid.).

Any new kind of venture requires experimentation, and brewpubs are no exception (Weiler, 2000). In Pennsylvania, the period between 2016 and 2018 saw a burst of newly permitted brewpub locations, with exactly 50 new locations being permitted over those three years (PLCB, 2026). This period roughly coincides with the period of greatest growth in the volume of reviews posted to Google Maps—the most popular review platform for information seekers (BrightLocal, 2023). Entrepreneurs can learn from the information available on their competitors' Google profiles, but they can also engage their own experienced customers for some signal of how to improve. Can Google reviewers have helped mobilize brewpub ideas, as 'cross-pollinators' of the brewpub landscape?

Google brewpub guides

Browsing the reviews of brewpubs on Google Maps, you may come across a review which indicates that the reviewer has reviewed several other brewpubs. The implication is that this user is developing some expertise in the brewpub experience and, therefore, their opinions should be weighted according. The upper left image of Figure 2 shows an example. When I visit this user's profile, the bio states "Love to travel to breweries". And I can peruse the map of all amenities the use has reviewed. With 249 reviews and ratings, a large share of which are for bars, breweries, and brewpubs, dotting the expanse from Philadelphia to Pittsburgh, this user is a good candidate for the status of 'cross-pollinator'.

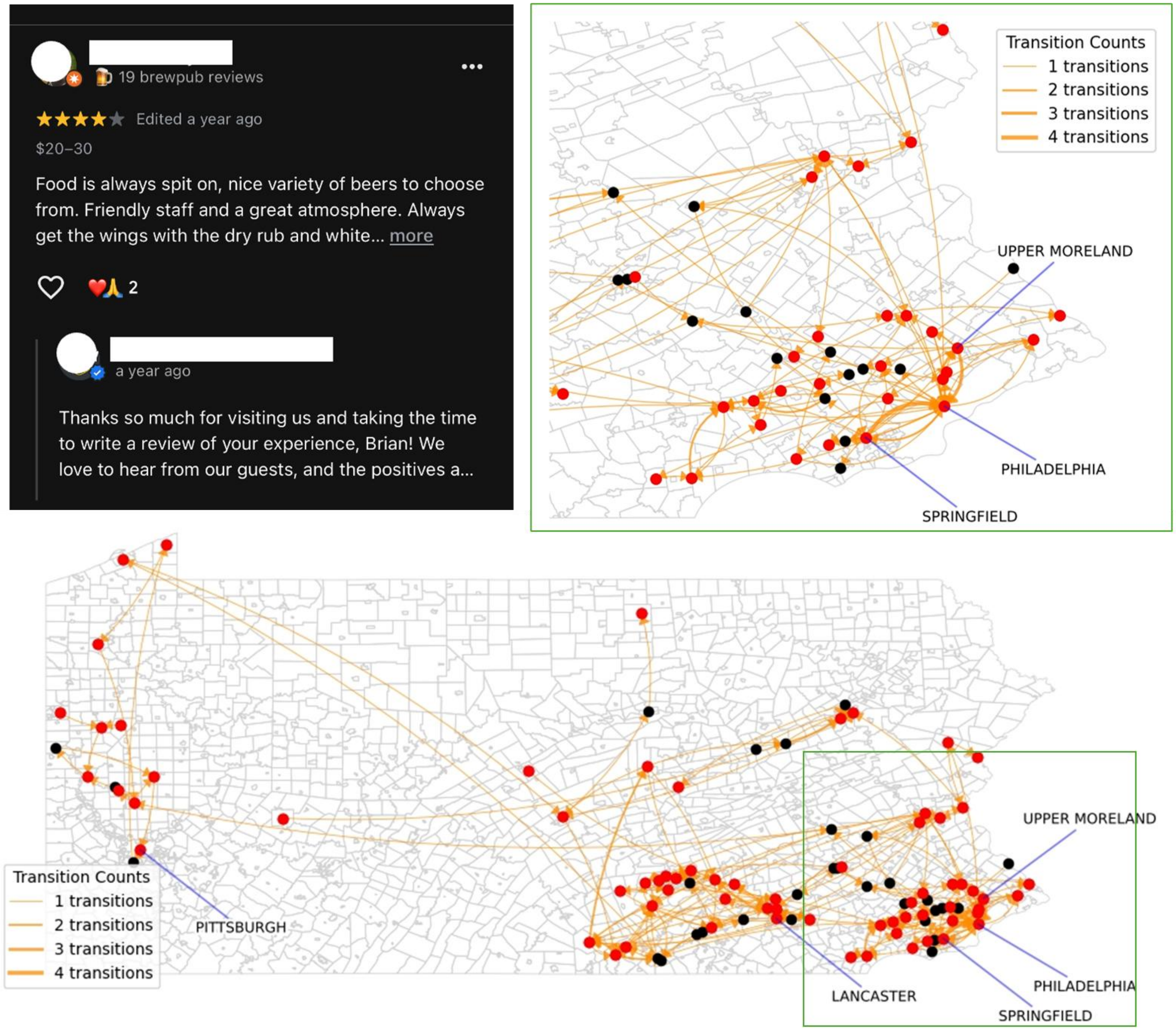

Figure 2: Upper left) screenshot of Google review of a Pennsylvania brewpub. Lower) map of intermunicipal transitions for sample of users who have at least one non-five-star brewpub review in Pennsylvania which has been responded to, and which contains a suggestion from improvement. Municipalities containing at least one of such reviews is highlighted red. Upper left) Zoom-in of map.

Using Google data collected in August 2021 by the Mcauley Lab (Li et at., 2022; Yan et al., 2023), we can identify other such candidates who have operate across the state. In the Pennsylvania dataset, there are 248 unique business locations with "brewpub" included in the category field, and they are distributed across 154 different Pennsylvania municipalities, with the highest concentrations in Philadelphia (26 locations), followed by Pittsburgh (12 locations) and Lancaster (8 locations). The user with the most brewpub reviews (47) in the dataset travelled across 40 different municipalities to experience these brewpubs.

Brewpub sentiment flows

Consider that each brewpub experience reviewed by a user is a conditioned on the prior brewpub experiences they have reviewed. That is, the previous brewpub experiences the user took the time to peruse, pursue, and review are remembered as a dynamic standard for judging their next brewpub experience, an idea consonant with sequential dependency theories (Vinson et al., 2019). In that case, every time a user travels across municipal lines, from one brewpub to another, this represents a potential flow of brewpub sentiment from one municipality to another.

To capture the potential brewpub sentiment flows mobilized by Google reviewers across Pennsylvania, we can fit user review paths into transition matrix model. For example, if Harry had the following brewpub locations [PHILADELPHIA ↣ LITITZ ↣ PHILADELPHIA ↣ LANCASTER], and Sally had [LITITZ ↣ PHILADELPHIA ↣ LANCASTER ↣ LANCASTER], then the sentiment flow potential from Philadelphia to Lititz was 1, from Lititz to Philadelphia was 2, and from Philadelphia to Lancaster was 2. Completing this operation for all users who reviewed two or more brewpubs, we can identify the highest potentials. We find that, out of 12,943 potentials, 256 (2%) were back and forth between Philadelphia and nearby Springfield, and 247 (1.9%) were back and forth between Philadelphia and nearby Upper Moreland. Given the concentration of brewpubs in Philadelphia, it is somewhat expected for these to be among the highest.

But these percentages are still relatively low. The broader map shows a weakly connected network of transitions stretching from east to west Pennsylvania, all of which are tied together by a single platform. The 36 users who posted brewpub reviews in 10 or more municipalities completed 447 directed intermunicipal transitions, connecting 122 municipalities (79% of the possible 154) with their brewpub reviews alone, also stretching from east to west in a weakly connected graph (see Figure C-1 in Supplemental Material). These users are potential super-mobilizers of brewpub sentiment, as their judgements connect many places over a vast landscape.

Brewpub pollinations

So much for the potential flow of sentiments, what about the actual communication of improvement suggestions? Using the present dataset, the best confirmation that communication has been established is if the user's review has been responded to by the owner (see upper left image of Figure 2). Users may travel around and share opinions, but it is some positive sign of reception if these opinions are publicly responded to. Some responses express things like "Your insights are crucial as we continuously strive to enhance our guest experience" (responder in dataset), and "We're always striving to improve and your comments will certainly help us do that" (responder in dataset). In the Pennsylvania dataset, 19% (10,185) of the 53,719 brewpub reviews with written text received owner responses. We want to know which responded-to reviews likely contain explicit suggestions of business improvements. We consider only those reviews with a non-five-star rating, as these suggest room for improvement more obviously, leaving us with 4,386 (43.1%) of the responded-to reviews. Although 5-star reviews, and reviews which have not been responded to, may still communicate useful feedback, we use the more stringent criteria to produce a manageable sample for LLM-ensemble classification (Schoenegger et al., 2024).

The method used includes, first, using GPT-4o-mini and Llama3 models to independently label the reviews either YES, NO, or MAYBE, regarding whether they include explicit suggestions for improvement. If these two models agreed on a label, then we accept it. If they disagreed, however, those reviews were given to Claude-sonnet-4-5 to complete the same labelling task. If this third model did not settle disagreements in the direction of either of the previous two, then the review was labelled the medial value MAYBE. Section C of the Supplement Material provides more details on this process. The outcome is 747 (17.0%) YES, 957 (21.8%) MAYBE, and 2,682 (61.1%) NO. Suggestions include things like prompter bar service, more careful attention to food order alterations, things that could be renovated to improve ambiance, and even how to present information on the Google profile.

Figure 2 shows a map of all the intermunicipal transitions made by any user with at least one review labelled YES. Municipalities with at least one brewpub review labelled YES are highlighted in red. As shown, even with these strict criteria of selection, a geographically-broad collection of brewpub locations was seeded with suggestions through the platform. Given that most users in the dataset have only one brewpub review, most of the reviews labelled YES are written by users who have no prior Pennsylvania brewpub reviews leading up to it. However, a respectable 12% (92) of non-five-star, responded-to, suggestion-providing are preceded by at least one Pennsylvania brewpub review by the same users.

Discussion

In this short exploratory study, we approached the GP as a surface which allows the flow of sentiments and suggestions across broad amenity landscapes. While most users would have posted very few reviews, including maybe one brewpub review, some users were found to have reviewed multiple brewpubs across many different Pennsylvania municipalities. Further, through selective LLM labelling, we identified a non-negligible number of exchanges between brewpub reviewers and providers around suggestions for improvement. Whereas conventional urban policy mobility literature centers professionals or institutionalized experts and their inter-municipal exchanges, in this exploration, we see how GP reviewers (especially enthusiastic ones who travel broadly) may serve as important mobilizers of place-based ideas as well.

For convenience, this study only included Pennsylvania brewpubs, although neighboring states like New Jersey have their own distribution of brewpubs which some users in the study likely reviewed. However, more significantly, there are some things this exploratory study left relatively implicit. A unique affordance of the P2P platform is that any user can pick up where another left off by simply reading their posts (Carr and Hayes, 2015). This is explicit, for example, when users refer to other users' reviews within their own review, which is not uncommon. Some users even explicitly compare establishments within their reviews, thereby drawing a thread for other users to follow. But the combination of explicit, implicit, and hidden influences that channel through the platform, and which comprise a regional flow, are difficult to measure. This exploratory study does not so much show that the attempt to improve such measurement is warranted (perhaps it is), but that the scale of the influence may be significant. It shows that, by facilitating communication and mutual accountability across locations and people, GPs may catalyze the learning process for location-based industries like brewpubs.

Conclusion

This paper asks how geosocial platforms (GP) like TripAdvisor, Yelp, and Google Local Guides intervene in processes associated urban policy mobility. It interpreted the convergence of the terms 'urban', 'policy', and 'mobility' somewhat loosely, to include any process by which mobile persons either 1) represent other people to influence the character of a place or 2) represent a place to influence the mobility decisions of other people. GPs are where these two forms of influence intersect and multiply, while emphasising the latter. Whereas the conventional approach to urban policy mobility centers territorial administrations and mobile policy professionals, GPs shift the emphasis towards global computational infrastructures and their collective users.

This paper presented three specific ways to approach GPs and understand their place-based influences. This was done through three short empirical studies, which demonstrated how geosocial platforms may function as 1) tools for engaging public feedback around place-based interventions; 2) technologies which embed their own kind of urban policy by rewarding certain travel behaviours; and 3) accountability systems which facilitate the flow of sentiments and ideas across landscapes.

Regarding physical space, the three studies operationalize three different geographic scales: from geolocation to metro area to statewide. However, regarding cultural theme, the three studies scale almost in the inverse direction: from cosmopolitan tourism to 'scoreboard urbanism' to place-based entrepreneurship. The intersection of scales demonstrated across the studies reflects a general regime of influence, institutionalized by GPs, which largely disregards parochial territorial concerns. However, it is by studying this disregard that urban policy professionals can learn to appropriate it.

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Supplementary Material

The data and code necessary to reproduce each of the three studies is published on GitHub (https://github.com/kjrm/Geosocial_Platforms_as_Urban_Policy_Mobility_Media-Replication.git).

Section A

This section unpacks methods pertaining to *Study 1: Platform as feedback tool*.

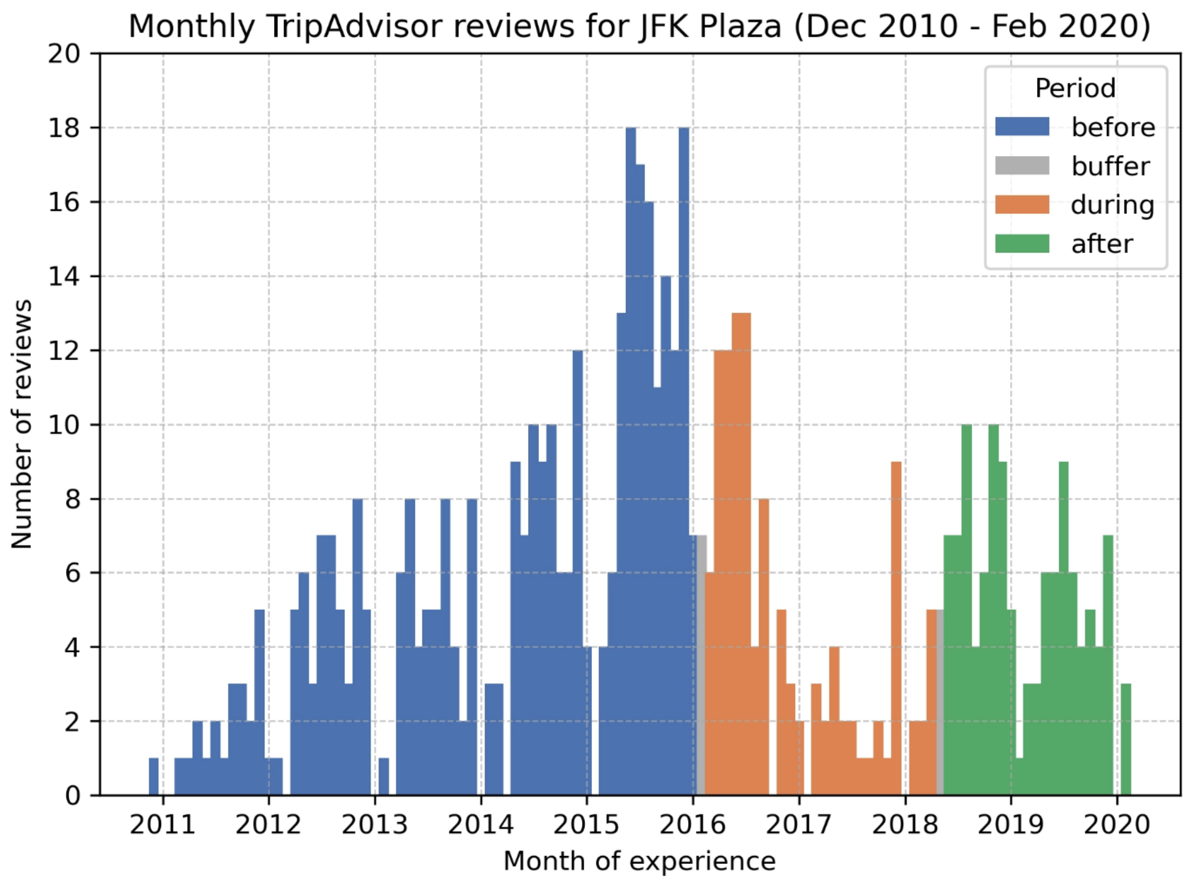


Figure A- 1: Monthly TripAdvisor posts for JFK Plaza, according to the month-year of experience specified in the post. Posted experiences begin in December 2010 and were collected up to the end of February 2020. The three study periods are highlighted, along with the buffer months for which posts were dropped.

Table A- 1: Descriptive statistics for posts included in Study 1

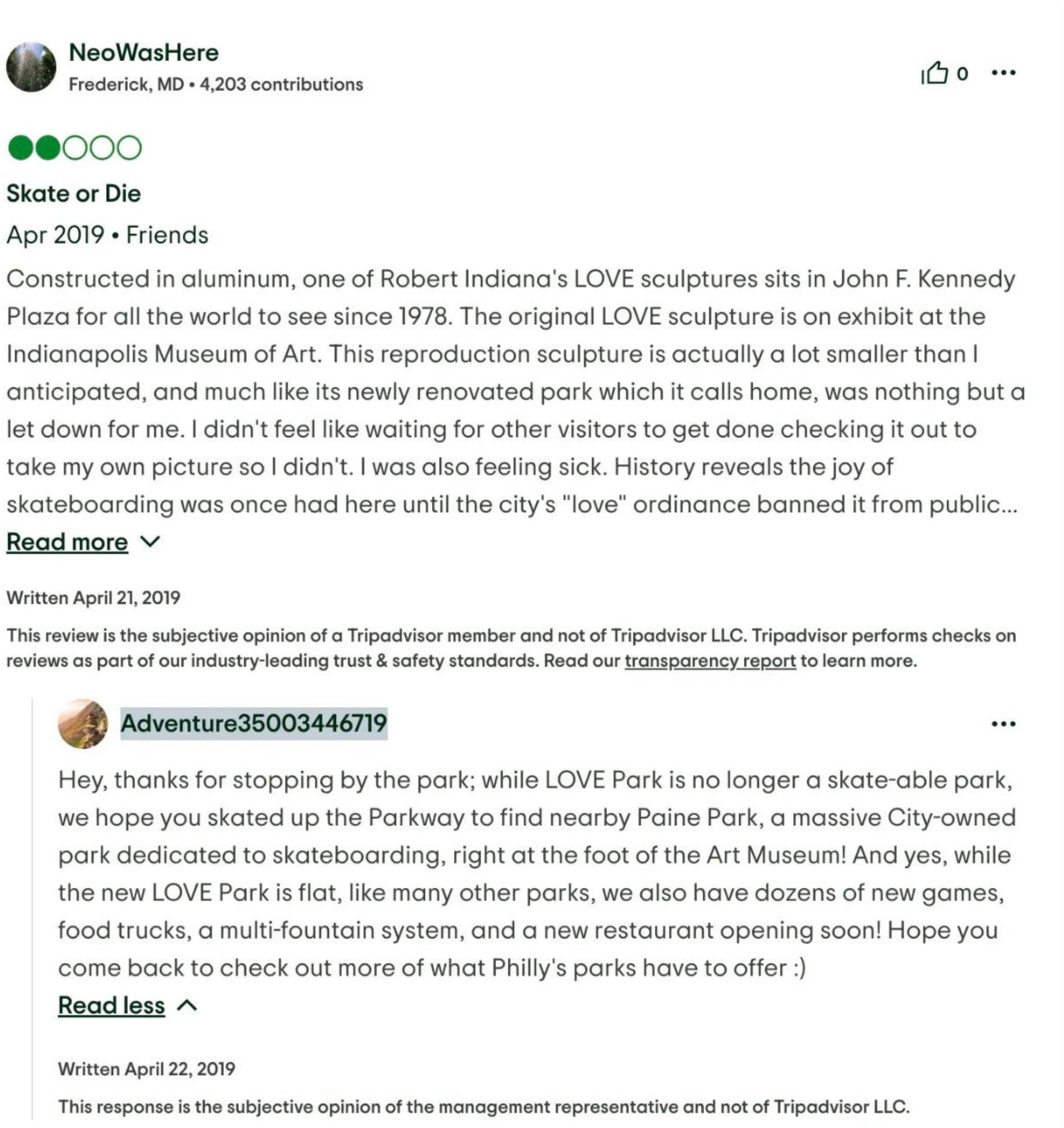
Rating (n = 568)	**before (n = 339)**	**during (n = 114)**	**after (n = 115)**
1	0.04	■0.11	0.03
2	0.07	0.09	0.06
3	■■■0.33	■■0.24	■■0.23
4	■■■0.34	■■■0.30	■■■0.34
5	■■0.22	■■0.27	■■■0.32
Party (n = 423)	**before (n = 243)**	**during (n = 95)**	**after (n = 85)**
Solo	■0.12	■0.15	■0.14
Friends	■■0.24	■0.19	■0.14
Couples	■■■0.32	■■0.29	■■■0.39
Family	■■0.26	■■■0.33	■■■0.31
Business	0.06	0.04	0.02
Photos (n = 568)	**before (n = 339)**	**during (n = 114)**	**after (n = 115)**
yes	■0.18	■■0.23	■■■0.37
Avg. no. of photos	0.29	0.71	0.97

BERTopic modeling

BERTopic modeling was performed for each of the three periods independently, using the same model parameters. The reason the same model is used across each of the periods is because we do not want to give differential weighting to the reviewers depending on the period their review is in. The model parameters were selected to produce an appropriate number of topics (15>n>3), while trying to keep the outlier share to an acceptable level (<30% for short texts), across each of the periods. Ultimately, however, these things meant to support topic interpretability. Testing different combinations of the two parameters *nearest neighbours* (how many neighbors to look at to define the "shape" of the data) and *minimum cluster size* (the minimum number of reviews required to form a distinct topic), we settled on values of 10 and 6, respectively. While this combination produced relatively high outlier shares, it seemed necessary not simply to induce a nontrivial number of topics (>3), but generally to boost the relevant signals across the reviews. The reality is that different reviewers express didn't levels of care in their reviews, and we most want to detect when there is agreement across those who care relatively strongly. Table A-2 summarizes the results.

Table A- 2: Topic modeling results

Topic	Count	Topic terms
before reconstruction (outlier share = 23%)		
1-1	**62**	park_philadelphia_philly_love park
1-2	**35**	christmas_village_christmas village_vendors
1-3	**30**	fountain_water_area_sculpture
1-4	**27**	park_picture_statue_sculpture
1-5	**22**	picture_pictures_homeless_nice
1-6	**16**	skateboarders_homeless_area_homeless people
1-7	**11**	homeless_homeless people_scary_park
1-8	**11**	christmas_village_park_christmas village
1-9	**11**	brotherly love_brotherly_city brotherly_city
1-10	**10**	kids_sign_picture_shot
1-11	**9**	sign_smaller_smaller imagined_imagined
1-12	**9**	took_anniversary_photos_park
1-13	**8**	love park_admire_park_fun
during reconstruction (outlier share = 31%)		
2-1	**30**	sign_love sign_small_really
2-2	**20**	taking_photo_spot_place
2-3	**20**	statue_love park_love statue_plaza
2-4	**9**	philly_center city_center_easy
after reconstruction (outlier share = 23%)		
3-1	**42**	love park_line_picture_sculpture
3-2	**14**	christmas_village_vendors_christmas village
3-3	**13**	sign_picture_parking_love sign
3-4	**12**	sign_love sign_philly_pictures love
3-5	**7**	philly_kids_love park_clean



NeoWasHere
Frederick, MD • 4,203 contributions

0

Skate or Die

Apr 2019 • Friends

Constructed in aluminum, one of Robert Indiana's LOVE sculptures sits in John F. Kennedy Plaza for all the world to see since 1978. The original LOVE sculpture is on exhibit at the Indianapolis Museum of Art. This reproduction sculpture is actually a lot smaller than I anticipated, and much like its newly renovated park which it calls home, was nothing but a let down for me. I didn't feel like waiting for other visitors to get done checking it out to take my own picture so I didn't. I was also feeling sick. History reveals the joy of skateboarding was once had here until the city's "love" ordinance banned it from public...

Read more

Written April 21, 2019

This review is the subjective opinion of a Tripadvisor member and not of Tripadvisor LLC. Tripadvisor performs checks on reviews as part of our industry-leading trust & safety standards. Read our transparency report to learn more.

Adventure35003446719

Hey, thanks for stopping by the park; while LOVE Park is no longer a skate-able park, we hope you skated up the Parkway to find nearby Paine Park, a massive City-owned park dedicated to skateboarding, right at the foot of the Art Museum! And yes, while the new LOVE Park is flat, like many other parks, we also have dozens of new games, food trucks, a multi-fountain system, and a new restaurant opening soon! Hope you come back to check out more of what Philly's parks have to offer :)

Read less

Written April 22, 2019

This response is the subjective opinion of the management representative and not of Tripadvisor LLC.

Figure A- 2: Screenshot of the only TripAdvisor review of JFK Plaza seen to have received a response, in the after-reconstruction period.

Section B

This section unpacks methods pertaining to *Study 2: Platform as policy implement*.

Identifying scouts from the distribution of first-reviews

Scouts should have a far higher first-reviews than otherwise expected. If users just happened to accumulate first-reviews the more they review, by chance, and there is no competition around first-reviews, then we would expect the distribution of first-reviews to be approximately binomial (null expectation). We can get an estimate of this null expectation by treating each review as a Bernoulli trial where there is a fixed probability ($\hat{p}$) of getting a first-review. That probability is the overall rate of first-reviews within the entire pool of included user reviews.

f_i is the observed number of first-reviews for user *i,*

n_i is the observed total number of reviews for user *i,*

$\hat{p} = \frac{\sum f_i}{\sum n_i} = \frac{7{,}408}{329{,}985} = 0.022$ is the overall, or pooled, rate of first-reviews.

Therefore, under null conditions, the expected number of first-reviews for each user is

$\hat{f}_i = n_i \hat{p}$

The Pearson residual standardizes the difference between observed and expected for each user

$r_i = \frac{f_i - n_i \hat{p}}{\sqrt{n_i \hat{p}(1-\hat{p})}}$

We will create a (right) tail plot to compare observed, versus expected, numbers of individuals passing different residual thresholds. We take a range of threshold values 0 to 5 in 0.05 increments. At each threshold, *t*, we count the number of users observed to have a Pearson residual greater than or equal to it. We can compare this to the number of users expected to pass this threshold, expressed as

$E(t) = \sum_i^N P(F_i > f^*(n_i, t))$

where $k^*(n_i, t)$ is the smallest integer count that would produce a residual of at least t for user i, and each probability is evaluated exactly from the binominal distribution with parameters n_i and $\hat{p}$.

Divergence between these two curves in the right tail indicates systematic excess of high-residual users beyond what chance would produce. Figure B-1 seems to show that. The two lines cross once, with observed tail count persistently higher than the expected past a threshold of around 1. This skewing supports the notion of a selective competition among users, where some make the effort to be first reviewer and succeed, while other don't and 'lose out'. A safe threshold of 2.5 is used to distinguish scouts, as this is around where we see the greatest excess of users above chance.

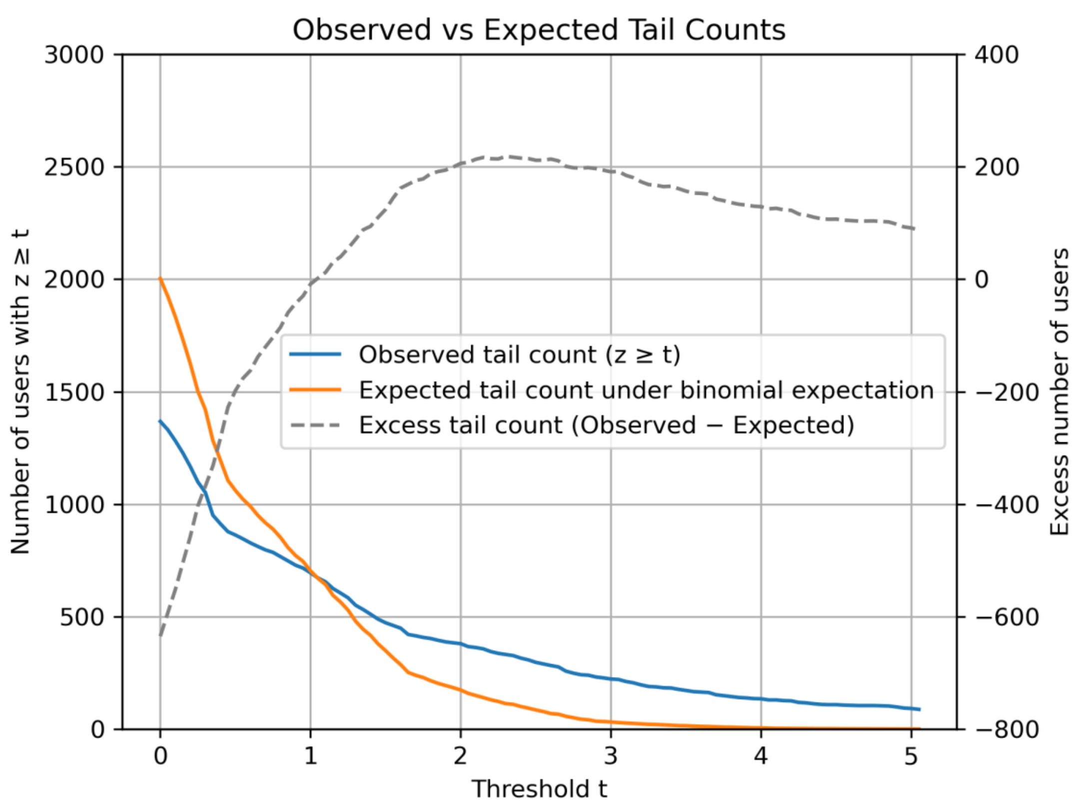


Figure B- 1: Tail count analysis to identify Yelp scouts in Philadelphia metro area

Table B- 1: Sensitivity analysis

Parameters				
Outer radius	50	60	70	75
Inner radius	12	15	18	20
Min. user reviews	25	30	32	35
threshold	2.5	2.5	2.25	2.5
pairs	244	227	265	202
Results				
Radius of gyration				
scout > control	0.631	0.652	0.619	0.639
Binomial p	0.000025	0.000003	0.000065	0.000050
Total path length				
scout > control	0.635	0.678	0.638	
Binomial p	0.000014	0.000000	0.000004	0.000008

Section C

This section unpacks methods pertaining to *Study 3: Platform as facilitator of sentiment and idea flow*.

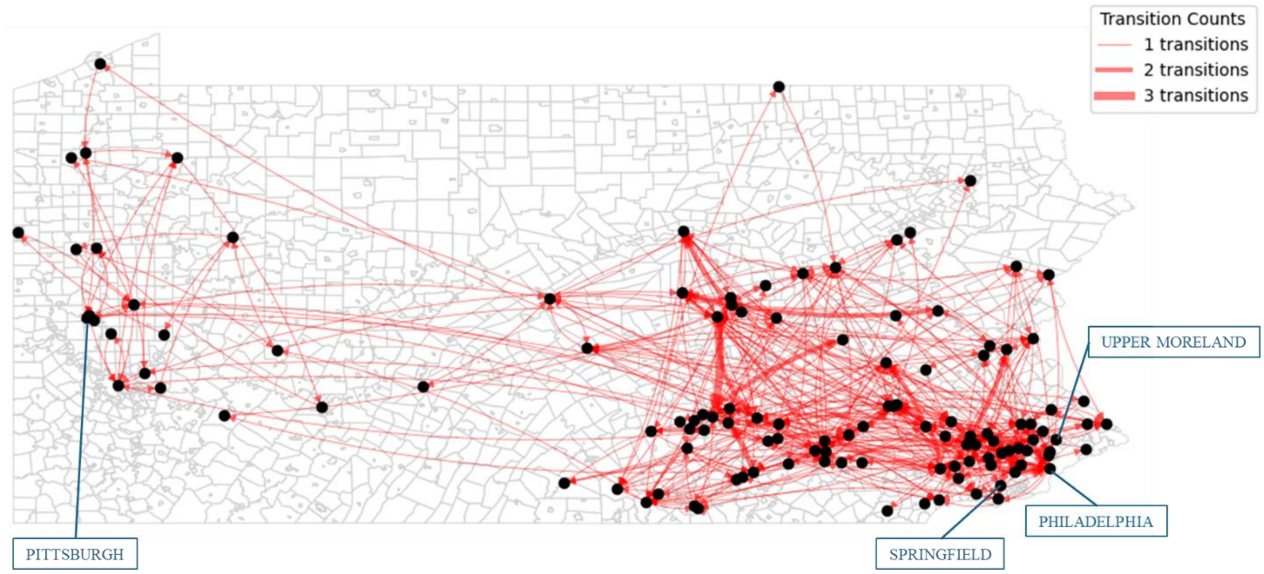


Figure C- 1: Inter-municipal transitions for brewpub reviewers with 10 or more unique municipalities review in. The transitions comprise a weakly connected graph, with 122 nodes (79% of the possible 154 municipality locations for Pennsylvania brewpubs).

Using LLMs to label reviews

To find which of the selected reviews contain explicit suggestion for improvement, three different large language models (LLM) were used. Each model was employed using zero-shot prompting, including structured examples (see Figure C-2). First, GPT-4o-mini, and Llama3, were each independently asked to label each of the selected reviews, either YES, MAYBE, or NO. This resulted in an agreement rate of 74.8%. The 25.2% of reviews for which there was disagreement between the two models were fed to Claude-sonnet-4-5 to complete the same labelling task, to try to settle the disagreement toward one side or the other. This produced 840 settled disagreements. The remaining 299 disagreements that were not settled—i.e. reviews that were labelled some permutation including YES, MAYBE, and NO across the three models—were hard-labelled MAYBE. Figure C-2 shows the structure of the process, and include the system prompt that was used for each labelling task. The final outcome is 747 (17.0%) YES, 957 (21.8%) MAYBE, and 2,682 (61.1%) NO.

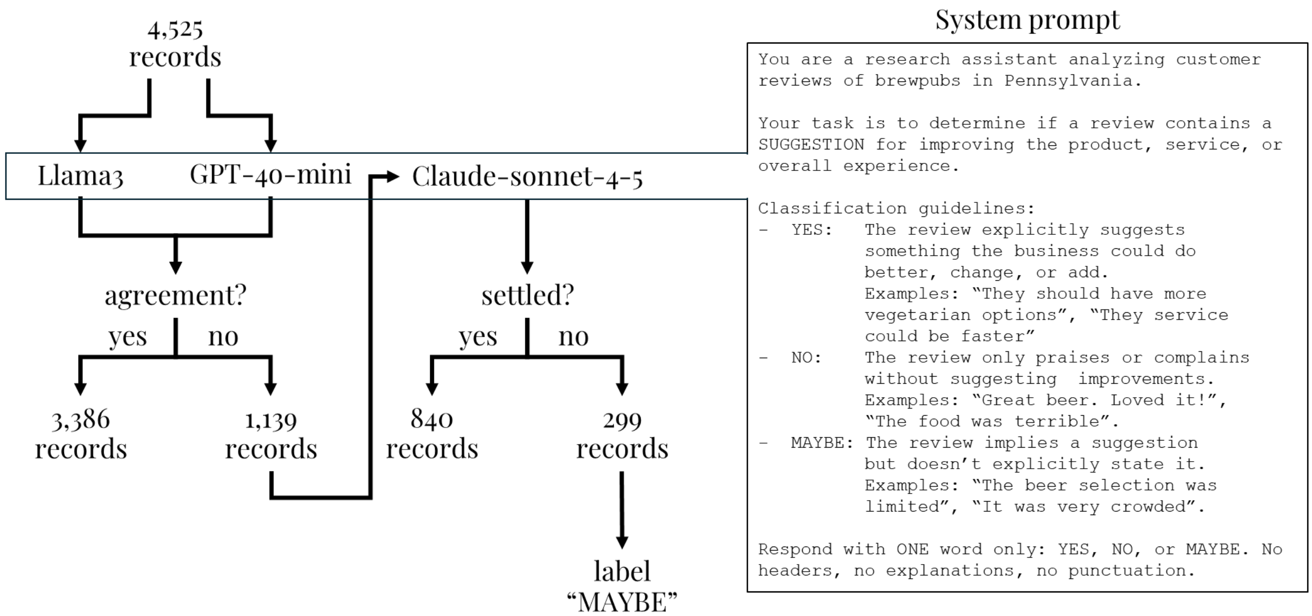


Figure C- 2: Overview of the review labelling process